

Federated Semi-Supervised Medical Image Segmentation via Prototype-Based Pseudo-Labeling and Contrastive Learning



Course: Image Processing
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Problem Statement

Medical image segmentation models require large amounts of annotated data for high performance. However:

- Medical data is distributed across multiple hospitals
- Due to privacy regulations, data cannot be shared
- Many sites only have unlabeled medical images

Traditional:

- Federated Learning (FL) assumes all clients have labeled data
- Semi-supervised learning assumes labeled & unlabeled data exist at same location

Real-world challenge:

Some clients have labeled data, while many others only have unlabeled data — and data cannot be shared.

Objective

To develop a Federated Semi-Supervised Segmentation (FSSS) framework that:

- Effectively utilizes limited labeled data
- Exploits abundant unlabeled data
- Maintains data privacy
- Improves segmentation performance in medical imaging

Specifically, the proposed method aims to:

- Reduce bias in pseudo-label generation
- Enhance feature discrimination in unlabeled clients
- Dynamically aggregate models based on training reliability
- Improve global model convergence under non-IID data

Methodology Overview

The proposed framework utilizes three key novel strategies to enhance learning in a semi-supervised federated environment

1. **Pseudo-Labeling Debias** (PLD): Uses a "Labeled-Aggregated Model" to verify and filter pseudo-labels.
2. **Prototypical Contrastive Learning** (PCL): Uses feature centroids from labeled data to guide unlabeled models.
3. **Dynamic Model Aggregation** (DMA): Adjusts server weights based on model consistency rather than just data size.

1. Prototype-Based Pseudo-Labeling Debias

Problem:

Pseudo-labels generated by the global model may be unreliable due to:

- Non-IID data distribution
- Weak training at unlabeled clients

Solution:

- Extract prototypes (feature centroids) from labeled clients
- Compute similarity between labeled and unlabeled clients
- Build a Labeled-Aggregated Model
- Generate pseudo-labels only when:
 - Global model prediction is confident
 - Labeled-aggregated model agrees
 - Prediction entropy is low

2. Prototypical Contrastive Learning

Problem:

Unlabeled clients lack strong discriminative features.

Solution:

- Transfer image-level prototypes from labeled clients
- Apply contrastive learning:
 - Bring similar class prototypes closer
 - Push foreground and background apart

Loss used:

- Dice Loss (for segmentation): Measures overlap between predicted mask and ground truth mask.
- InfoNCE Loss (for contrastive learning): Pulls similar features closer and Pushes different features apart.

3. Dynamic Model Aggregation (DMA)

Problem:

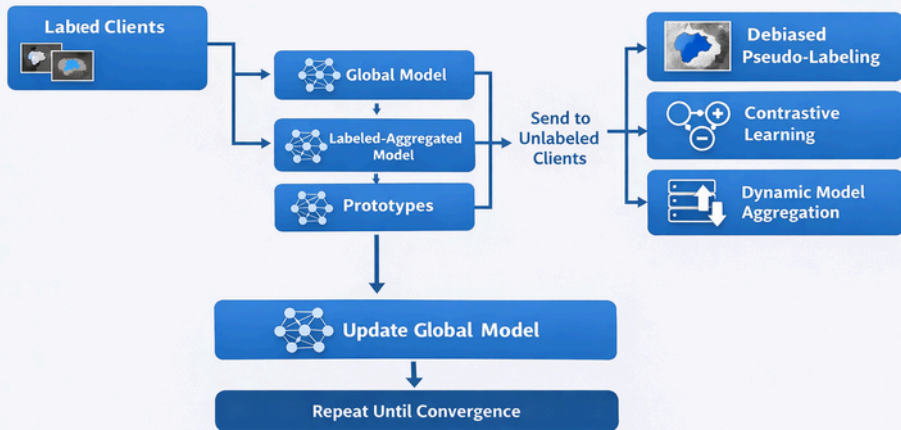
Unlabeled models are unstable in early federated rounds.

Traditional Federated Averaging gives weight based only on data size.

Solution:

- Assign optimization coefficient:
 - Labeled clients $\rightarrow$ coefficient = 1
 - Unlabeled clients $\rightarrow$ based on prediction consistency
- Aggregate models using:
 $\text{Weight} \propto (\text{Optimization degree} \times \text{Dataset size})$

System Architecture



Training Setup

Dataset: QaTa-COV19

- Medical chest X-ray segmentation dataset
- Images resized to 224×224
- Binary segmentation masks

Model Architecture: We adopt a U-Net architecture with a ResNet-34 encoder pretrained on ImageNet.

- Encoder: ResNet-34 backbone
 - Decoder: Transposed convolution layers with skip connections
 - Output: 1-channel segmentation mask (sigmoid activation)
 - Input resolution: 224×224
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- Split: 4 federated clients => 2 labeled clients 2 unlabeled clients
 - Training: 15 federated rounds, Adam optimizer, Learning rate = $1e-4$, Batch size = 8

Results

Baseline Method

Supervised Federated Learning

Model: ResNet34 + Dice Loss

Setup: 2 Labeled Clients

Dice Score: 0.8123

Proposed Method

FSSS + Contrastive Learning

Model: ResNet34 + Prototype Learning +
Contrastive Loss

Setup: 2 Labeled and 2 Unlabeled clients

Dice Score: 0.8422

Performance Improvement

$$\Delta \text{Dice} = 0.8422 - 0.8123 = 0.0299$$

Absolute Improvement: +0.0299

Relative Improvement: $\approx 3.0\%$ increase

Future Scope

- Clustering Integration: Implement K-means to handle multi-modal feature distributions within the lung segments.
- Temporal Memory: Integrate an EMA-based prototype bank to stabilize training at unlabeled sites.
- Techniques to narrow the "client drift" and distribution gap specifically for medical imaging.
- Limited to 15 rounds due to hardware constraints, though still demonstrating a ~3% Dice score improvement over the baseline.
- Explore ways to reduce the communication overhead, as unlabeled clients currently require double the bandwidth to download both global and labeled-aggregated models.

Conclusion

This work successfully implements a prototype-based federated semi-supervised segmentation framework with contrastive learning using a ResNet-34 backbone.

The proposed method achieves a **Dice score of 0.8422**, outperforming the supervised baseline by 3.0%, demonstrating the effectiveness of prototype-guided contrastive representation learning in federated medical image segmentation.

THANK YOU